Simple if and if-else Statements in Java

In Java, conditional statements allow us to perform different actions based on different conditions. The `if` and `if-else` statements are used to control the flow of the program based on boolean expressions (true or false).

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1. The `if` Statement

The `if` statement is used to execute a block of code only if a specified condition is true. If the condition evaluates to `false`, the block of code inside the `if` statement is skipped.

Syntax:

```
if (condition) {
    // Code to be executed if the condition is true
}
```

- **Condition:** A boolean expression that evaluates to `true` or `false`.

Example:

```
public class Main {
    public static void main(String[] args) {
        int number = 10;

        if (number > 5) {
            System.out.println("Number is greater than 5");
        }
    }
}
```

- **Explanation:**
  - Here, the condition `number > 5` evaluates to `true`, so the statement inside the `if` block is executed, printing `"Number is greater than 5"`.
  - If the condition were false (e.g., if `number` was 3), the message would not be printed.

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2. The `if-else` Statement

The `if-else` statement allows you to execute one block of code if the condition is true, and another block of code if the condition is false.

Syntax:

```
if (condition) {
    // Code to be executed if the condition is true
} else {
    // Code to be executed if the condition is false
}
```

- **Condition:** A boolean expression that is evaluated to `true` or `false`.

Example:

```
public class Main {
    public static void main(String[] args) {
        int number = 3;

        if (number > 5) {
            System.out.println("Number is greater than 5");
        } else {
            System.out.println("Number is less than or equal to 5");
        }
    }
}
```

- **Explanation:**
  - Here, the condition `number > 5` is `false` because `number` is 3. Therefore, the code inside the `else` block is executed, printing `"Number is less than or equal to 5"`.
  - If `number` was greater than 5, the message `"Number is greater than 5"` would have been printed.

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3. Nested `if` Statements

You can also nest `if` and `if-else` statements inside each other. This means placing one `if` or `if-else` inside another `if` or `else` block.

Example:

```
public class Main {
    public static void main(String[] args) {
```

```java
        int number = 10;

        if (number > 0) {
            if (number > 5) {
                System.out.println("Number is greater than 5");
            } else {
                System.out.println("Number is between 0 and 5");
            }
        } else {
            System.out.println("Number is less than or equal to 0");
        }
    }
}
```

- **Explanation:**
  - The first `if` checks if the `number` is greater than 0. Since the number is 10, the second `if` checks if it is greater than 5.
  - If both conditions are true, it prints `"Number is greater than 5"`.
